Digital Twin PoC – Happiest Health Challenge 2026

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- Project: fusing static EHR and real-time wearable data
- Goal: predict a glucose spike > 180 mg/dL 2 hours ahead

Problem and healthcare use case

- EHR data is static and infrequent; wearables are continuous
- Doctors rarely see both together
- Use case: pre-consult risk review of a virtual patient

Data streams (sandbox rules)

- Static: simulated EHR (age, BMI, HbA1c, diabetes, hypertension, genetic
- Dynamic: simulated hourly wearable/CGM over 14 days
- No real patient data; Synthea / MIMIC-IV can replace it at scale

Architecture

- Two streams -> feature fusion (7-day window) -> gradient-boosted model
- Output: risk score shown on a doctor dashboard
- See architecture_diagram.pdf

Model and results (held-out data)

- EHR only: AUC 0.981
- Wearable only: AUC 0.989
- Fused twin: AUC 0.988
- Positive cases ~1.5%, so AUC is the main metric

Honest findings and limitations

- Fused model did not beat wearable-only on synthetic data
- Likely cause: synthetic spikes tied to the same glucose signal
- Next: validate on realistic open datasets and calibrate risk

Outcome and next steps

- Working end-to-end prototype: data -> model -> dashboard
- Doctor dashboard concept showing top-risk virtual patients
- Next: real open datasets, precision/recall, clinical review